

DRAINAGE SYSTEM INSTALLATION I

ISCED UNIT CODE: 0732 251 03A

TVET CDACC UNIT CODE: CON/ CU/ PLM/CR /03 /5/ MA

UNIT DURATION: 100 Hours

Relationship to Occupational Standards

This unit addresses the Unit of Competency: **Install drainage system I.**

UNIT DESCRIPTION

This unit specifies the competencies required to install domestic drainage system. It involves Installing above ground drainage system, installing below ground drainage system, and maintaining drainage systems. It applies in the construction industry.

Summary of Learning Outcomes

S/No	Learning Outcomes	Duration (Hours)
1.	Install above ground drainage system	30
2.	Install below ground drainage system	30
3.	Interpret simple domestic drainage system drawings	20
4.	Maintaining drainage systems	20

Learning Outcomes, Content and Suggested Assessment Methods

Learning Outcome	Content	Suggested Assessment Methods
1. Install above ground drainage system	1.1 Safety Measures and Personal protective equipment 1.1.1 Hardhat/ Helmet 1.1.2 Safety gloves 1.1.3 Dustcoat / overall 1.1.4 Safety shoes / boots 1.1.5 Safety goggles/ face mask 1.1.6 Workshop SOP's, rules and regulations.	• Practical • Oral assessment • Observation • Portfolio of evidence • Written assessment

	1.2 Types of above ground drainage materials /accessories preparations 1.2.1 Waste pipes 1.2.2 Fittings 1.2.3 Bends 1.2.4 Tees 1.2.5 Couplings 1.2.6 Adapters 1.2.7 Cross tee 1.2.8 Pitcher tee 1.2.9 Inspection tee 1.2.10 Hangers and Supports 1.2.11 Pipe Hangers 1.2.12 Brackets 1.2.13 Clamps 1.2.14 Clips 1.2.2 Drainage Fixtures 1.2.15 Sinks (Kitchen, Bathroom) 1.2.16 W/C pan 1.2.17 Cistern 1.2.18 Bathtubs 1.2.19 Showers 1.2.20 Floor Drains 1.2.3 Seals and Gaskets 1.2.4 Rubber Gaskets 1.2.5 Silicone Sealants 1.2.6 Gratings and Traps 1.2.7 Drain Grates 1.2.8 P-traps 1.2.9 Vent Pipes 1.2.10 Accessories 1.2.11 Cleanouts 1.2.12 Rodders 1.3 Types of Drainage tools and equipment 1.3.1 Hacksaw 1.3.2 Trowel 1.3.3 Tape measure 1.3.4 Spirit level/ laser level 1.3.5 Steel float 1.3.6 Wooden float 1.3.7 Drilling machine 1.3.8 Screw drivers 1.3.9 Power extension cable 1.3.10 Mason hammer 1.3.11 Builders square 1.3.12 Plumbing snake	
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	1.3.13 Toilet plunger 1.3.14 Sump/water pump 1.3.15 Drainage Grates and Frames 1.4 Setting out drainage pipework 1.4.1 Positioning 1.4.2 Marking 1.5 Installation of drainage pipework. 1.5.1 Install hangers and supports 1.5.2 Cut pipes to length 1.5.3 Assemble pipe fittings 1.5.4 Fix pipes 1.5.5 Check alignment and slope 1.5.6 Seal joints and connections 1.5.7 Install drainage fixtures 1.6 Drainage systems functionality test 1.6.1 Visual Inspection 1.6.2 Water Flow Test 1.6.3 Pressure Test 1.6.4 Air Test 1.6.5 Backflow Test 1.6.6 Sump Pump Test 1.6.7 Drainage Fixture Test 1.6.8 Flow Rate Measurement 1.6.9 Check for Odors 1.7 Housekeeping 1.7.1 Care and maintenance 1.7.2 Storage	
2. Install below ground drainage system	2.1 Personal protective equipment 2.1.1 Hardhat/ Helmet 2.1.2 Safety gloves 2.1.3 Dustcoat / overall 2.1.4 Safety shoes / boots 2.1.5 Safety goggles/ face mask 2.1.6 Workshop SOP's, rules and regulations. 2.2 Types of below ground drainage materials /accessories 2.2.1 Waste pipes 2.2.2 Fittings 2.2.3 Bends 2.2.4 Tees 2.2.5 Couplings 2.2.6 Adapters 2.2.7 Cross tee 2.2.8 Pitcher tee 2.2.9 Inspection tee	• Practical • Oral assessment • Observation • Portfolio of evidence • Written assessment

	2.2.10 Hangers and Supports 2.2.11 Pipe Hangers 2.2.12 Brackets 2.2.13 Clamps 2.2.14 Clips 2.2.15 Drainage Fixtures	
	2.3 Types of drainage tools and equipment 2.3.1 Hacksaw 2.3.2 Trowel 2.3.3 Tape measure 2.3.4 Spirit level/ laser level 2.3.5 Steel float 2.3.6 Wooden float 2.3.7 Drilling machine 2.3.8 Screw drivers 2.3.9 Power extension cable 2.3.10 Mason hammer 2.3.11 Builders square 2.3.12 Plumbing snake 2.3.13 Toilet plunger 2.3.14 Sump/water pump 2.3.15 Drainage Grates and Frames 2.4 Preparation of Drainage trenches 2.4.1 Mark the Drainage Line 2.4.2 Excavate the Trench 2.4.3 Prepare the Trench Bottom 2.4.4 Setting out the drainage pipes 2.5 Installation of drainage pipework. 2.5.1 Set the Pipe 2.5.2 Install hangers and supports 2.5.3 Cut pipes to length 2.5.4 Assemble pipe fittings 2.5.5 Fix pipes 2.5.6 Check alignment and slope 2.5.7 Seal joints and connections 2.5.8 Install drainage fixtures 2.6 Construction of Drainage access points 2.6.1 Cleanouts 2.6.2 Inspection Chambers (Manholes) 2.6.3 Septic tanks 2.6.4 Bio digester 2.6.5 Access Wells 2.6.6 Catch Basins 2.6.7 Drains 2.6.8 Sump Pumps 2.6.9 Vent Pipes	

	2.6.10 Cleanout Fittings 2.6.11 Downspout Outlets 2.6.12 Traps 2.7 Drainage systems functionality tests 2.7.1 Visual Inspection 2.7.2 Water Flow Test 2.7.3 Pressure Test 2.7.4 Air Test 2.7.5 Backflow Test 2.7.6 Sump Pump Test 2.7.7 Drainage Fixture Test 2.7.8 Flow Rate Measurement 2.7.9 Check for Odors 2.8 Housekeeping 2.8.1 Care and maintenance 2.8.2 Storage	
3. Interpret simple domestic drainage system drawing	3.1 Working drawings 3.1.1 Pictorial 3.1.2 Line drawing 3.1.3 Freehand sketching 3.1.4 Scale drawings 3.2 Measurements 3.3 Symbols 3.4 Isometric pipework drawings 3.5 Interpreted working drawings.	• Practical • Oral assessment • Observation • Portfolio of evidence • Written assessment
4. Maintain drainage systems	4.1 Domestic drainage repair materials 4.1.1. PVC Pipes and Fittings 4.1.2. PVC Cement 4.1.3. Plumber's Putty 4.1.4. Teflon Tape 4.1.5. Rubber Gaskets and Washers 4.1.6. Drain Auger (Plumber's Snake) 4.1.7. Liquid Drain Cleaner 4.1.8. Pipe Insulation (foam or rubber) 4.1.9. Replacement Grates and Covers 4.1.10. Repair Clamps and Sleeves 4.1.11. Silicone Caulk 4.1.12. Hand Tools (pipe wrench, pliers, screwdrivers, hacksaw) 4.1.13. SOP's for drainage systems maintenance. 4.2 Notice for maintenance operations 4.3 Tools and equipment 4.3.1 Pipe Wrenches 4.3.2 Drain Auger (Plumber's Snake) 4.3.3 Plumber's Plunger	• Practical • Oral assessment • Observation • Portfolio of evidence • Written assessment

	4.3.4 Hacksaw 4.3.5 PVC Pipe Cutter 4.3.6 Channel Lock Pliers 4.3.7 Adjustable Wrench 4.3.8 Pipe Inspection Camera 4.3.9 Hand Trowel 4.3.10 Pipe Locator Tool 4.3.11 Shovel 4.3.12 Level 4.3.13 Drain Cleaning Machine (Electric Auger) 4.3.14 Heat Gun 4.3.15 Wet/Dry Vacuum 4.3.16 Measuring Tape 4.3.17 Safety Equipment (gloves, goggles, and mask) 4.3.18 Bucket 4.3.19 Teflon Tape 4.3.20 Pipe Sealant 4.4 Drainage pipework faults/ Blockages 4.5.1 Leaks 4.5.2 Corrosion 4.5.3 Root Intrusion 4.5.4 Pipe Misalignment 4.5.5 Bellied (Sagging) Pipes 4.5.6 Joint Failure 4.5.7 Cracks or Fractures 4.5.8 Backflow 4.5.9 Clogged Vent Pipes 4.5.10 Blockage 4.5.11 Pipe burst 4.5.12 Loss of trap seals 4.6 Housekeeping 4.7.1 Care and maintenance 4.7.2 Storage	
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Suggested Methods of Delivery

- Practical
- Direct Instruction
- Discussion
- Demonstration
- Trade projects
- Site visit

Recommended resources for 25 trainees

S/No.	Category	Resource Description/Specifications	Quantity	Recommended Ratio (Item : Trainee)
1.	Training Materials	Plumbing Principles and Practices Textbook	25	1 : 1
2.		Domestic Water Supply Installation Guide	25	1 : 1
3.		Technical Manuals (manufacturer's guides)	25	1 : 1
4.		Plumbing Workbook	25	1 : 1
5.		Sample Blueprints for Water Supply Layout	25	1 : 1
6.	Learning facilities and Infrastructure	Theory room (10m*8m)	1	1:25
7.		Workshop (18m*12m)	1	1:25
8.	Tools	Pipe Cutter	25	1 : 1
9.		Adjustable Wrench	25	1 : 1
10.		Basin Wrench	25	1 : 1
11.		Hacksaw	25	1 : 1
12.		Trowel	25	1 : 1
13.		Screwdrivers Set (Flathead and Phillips)	25 pieces	1 : 1
14.		Measuring Tape	25	1 : 1
15.		Plumber's Snake (Auger)	10	1 : 3
16.		Spirit level	25	1 : 1
17.		Steel/ Wooden float	25	1 : 1
18.		Mason hammer	25	1 : 1
19.		Drilling machine	5	1:5
20.		Builders square	25	1 : 1
21.		Power extension cable	5	1 : 5
22.	Equipment	Pipe Bending Machine	5	1 : 5
23.		Pressure Testing Kit	5	1 : 5
24.		Drilling machine	5	1 : 5
25.	Materials	PVC, PEX, Copper Pipes (1/2" and 3/4" diameter)	25 pieces	1 : 1
26.		Fittings (Elbows, Tees, Couplings)	25 pieces	1 : 1
27.		Valves (Gate, Ball, Check)	25 pieces	1 : 1
28.		Thread Seal Tape, adhesives and Pipe Joint Compound	25	1 : 1
29.		Pipe Insulation Material	25 pieces	1 : 1
30.		Screws (assorted)	sufficient	sufficient
31.		Adhesives	sufficient	sufficient

32.		Cement	sufficient	sufficient
33.		Sand	sufficient	sufficient
34.		Traps	sufficient	sufficient
35.		Caulking material	sufficient	sufficient
36.		Wall plugs	25 pieces	1 : 1
37.		Clips	25 pieces	1 : 1
38.	PPE's	Gloves, Goggles, Ear Protection, Masks, Safety shoes / boots	25 pieces	1 : 1
39.	Equipment	Water Pump Model (demonstration use)	1	1 : 25 (demonstration only)